

Nagaraju Vodnala

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SUMMARY

Senior Data Engineer with deep Azure and data lakehouse expertise, specializing in end-to-end cloud-native ETL/ELT pipeline delivery and cross-platform migrations. Achieved 99.9% data accuracy and accelerated compliance reporting by optimizing data validation, Delta Lake, and Spark jobs. Seeking to advance Pentair's analytics initiatives through Azure Data Factory and enterprise data platform innovation.

EDUCATION

University of Central Missouri | *Master's Degree, Computer Science*

MO, USA

CERTIFICATIONS

- AWS Data Engineer – Associate
- Microsoft Azure Data Engineer Associate
- SnowPro Core Certification

EXPERIENCE

Fifth Third Bank | *Senior Data Engineer*

Dec 2023 - Present

- Migrated legacy banking systems to a cloud-native Azure Data Lakehouse, delivering 10+ production ETL pipelines with ADF, Databricks, and Informatica, improving downstream analytics readiness and audit transparency.
- Designed, developed, and maintained Azure Data Factory pipelines for large-scale ELT workflows, integrating multiple sources and optimizing data extraction, loading, and transformation in a regulated environment.
- Integrated Azure Synapse with Informatica MDM, elevating customer data accuracy and enabling consistent 360° views across risk, credit, and servicing teams.
- Implemented and managed end-to-end data quality checks and profiling strategies, ensuring 99.9% data accuracy for regulatory and risk analytics.
- Partnered with data science teams to operationalize credit risk and fraud detection models in Databricks, enabling faster loan decisioning and real-time insights.
- Developed self-service Power BI models fed by real-time pipelines, improving portfolio visibility for business leaders and reducing reporting turnaround by 60%.
- Partnered with cross-functional business and analytics teams to gather requirements and translate them into robust data platform solutions supporting credit risk and fraud initiatives.
- Built GenAI-enabled summarization workflows using OpenAI APIs and Azure Functions, compressing lengthy compliance reports into actionable insights.
- Maintained clear documentation and data lineage maps to support audits, governance processes, and cross-team onboarding.

Progressive Corporation | *AWS Data Engineer*

Mar 2023 - Nov 2023

- Engineered a HIPAA-compliant claims data lake with AWS Glue, Lambda, and Lake Formation, consolidating 12 disparate data sources and improving claims traceability.
- Implemented streaming ingestion using Kafka and Kinesis, enabling near real-time processing for claims, payments, and event updates.
- Collaborated with data scientists to productionize ML models for fraud detection and risk scoring in SageMaker, reducing deployment cycles by 30%.
- Developed Flask- and Lambda-based risk scoring APIs, allowing customer-facing tools to consume ML outputs with low latency.
- Strengthened governance by enriching the Glue Data Catalog with lineage and access rules, ensuring full HIPAA compliance during audits.
- Enhanced Redshift performance through partitioning, sort keys, and materialized views-cutting analytics query times by 60%.
- Automated infrastructure with Terraform and CloudFormation, standardizing deployments and reducing manual configuration drift.
- Set up proactive monitoring using CloudWatch and Prometheus, ensuring 99.9% reliability across production ETL workloads.
- Created architecture diagrams and workflow documentation to improve cross-team alignment and operational understanding.

IBM India Pvt. Ltd | *Data Engineer*

Sep 2020 - Sep 2022

- Migrated a large-scale telecom warehouse to GCP BigQuery, designing ELT pipelines via Dataflow, Dataproc, and Pub/Sub to process 5TB+ of daily operational data reliably.
- Integrated Informatica MDM with GCP Data Catalog, improving customer identity resolution and reducing duplicate records by 85%.
- Developed near real-time anomaly detection using Apache Beam and BigQuery, reducing incident identification time from hours to minutes.
- Partnered with ML teams to deploy churn and network anomaly models via Vertex AI, improving retention insights by 18%.
- Optimized BigQuery spend through partitioning, clustering, and caching strategies, reducing monthly warehouse costs by 30%.
- Automated orchestration using Cloud Composer (Airflow), improving reliability of complex multi-stage pipelines.
- Delivered executive reporting through Looker and Tableau, surfacing KPIs such as usage insights, churn risk, and service degradation.
- Built GenAI-aided summarization workflows using Hugging Face and Vertex AI to streamline operational reporting.
- Set up full-stack monitoring with Stackdriver to meet strict SLA requirements and support rapid incident response.

Micro Labs Limited | *Data Engineer*

Jun 2019 - Aug 2020

- Designed ETL workflows integrating market feeds, trading systems, and internal platforms into a unified analytical store, strengthening data consistency and availability.
- Implemented ingestion pipelines using APIs, batch scheduling, and Kafka streaming to reduce data integration errors and latency.
- Optimized relational models and indexing strategies to improve application performance and accelerate analytics use cases.
- Developed dashboards in Tableau, Power BI, and Looker to support commercial teams with actionable insights.
- Established monitoring and alerting via open-source tools, improving ETL reliability and reducing downtime.
- Documented data mappings, quality rules, and end-to-end workflows using Confluence and PostgreSQL, enabling consistent governance across teams and reducing data-related issues during releases.

TECHNICAL SKILLS

- **Cloud Platforms:** Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP)
- **Developer Tools:** Git, Docker, Jenkins, TravisCI, VS Code, PyCharm, IntelliJ, Eclipse
- **Business Intelligence:** Power BI, Tableau, Looker, QuickSight
- **Data Engineering & Integration:** Azure Data Factory, ETL/ELT Pipelines, SSIS, Informatica PowerCenter, Informatica MDM, Talend, dbt, Apache NiFi, Kafka, Airflow
- **Databases & Warehousing:** SQL Server, PostgreSQL, Snowflake, MySQL, BigQuery, Oracle, Redshift, Hive
- **Programming Languages:** Python, SQL, Scala, Java, C/C++, R
- **Data Quality & Profiling:** Data Profiling, Data Validation, Quality Assurance
- **Frameworks & Technologies:** Apache Spark, PySpark, Hive, Flask, FastAPI